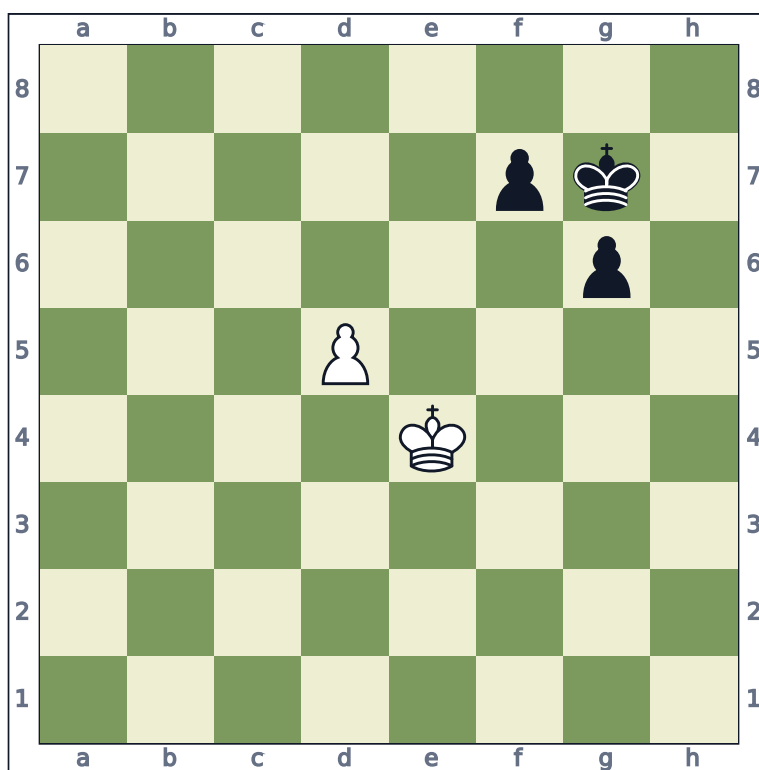


ENDGAME TECHNIQUE

Calculate Pawn Races Exactly

Count moves to promotion, checks, captures, and king entry rather than relying on visual intuition.



Illustrative endgame position - analyze the idea, not a memorized move

WHY THIS PRINCIPLE EXISTS

Pawn races are concrete. Promotion with check, a diagonal king route, or one reserve tempo can reverse the result.

THE MECHANISM

Why it works

This principle is not magic and it is not a commandment. It is a compressed prediction about conversion and tempo. The useful question is: what changes in the position when the principle is followed?

DIRECT CAUSAL MECHANISM

Pawn races are concrete. Promotion with check, a diagonal king route, or one reserve tempo can reverse the result.

BOARD GEOMETRY

With fewer pieces, king routes, promotion squares, checking distance, and pawn shape become easier to calculate exactly.

OPTION QUALITY

Each exchange can remove counterplay or accidentally erase the winning mechanism.

TIME AND TEMPO

Single tempi matter because promotion races, opposition, and zugzwang are often exact.

WHAT IT PREVENTS

It prevents a recurring class of errors by making a hidden cost visible before the move is played. The principle is valuable because that error pattern appears across many different openings and positions.

RECOGNITION

When to apply it

A principle becomes practical only when you can recognize its trigger. Use the signs below as evidence, then verify the exact position.

- Signal 1: Passed pawns on both wings.
- Signal 2: kings cannot stop everything.
- Signal 3: exchanges lead to a race..

FOUR-STEP APPLICATION PROTOCOL

1. Notice the trigger without choosing a move yet. 2. Generate at least two candidates, including one that follows the principle. 3. Calculate the opponent's strongest forcing reply. 4. Compare the resulting positions and apply the rule only if the concrete result supports it.

CONVERT IT INTO A BOARD QUESTION

Ask: 'Does this position give me a reason to use the following rule - Count moves to promotion, checks, captures, and king entry rather than relying on visual intuition.' Then ask the balancing question: 'What tactical or strategic detail could make this position an exception?'

EXCEPTIONS AND CONFLICTS

Where the slogan breaks

Chess strength comes partly from knowing when two good principles conflict. The better player does not repeat the louder slogan; they rank the actual consequences.

IMPORTANT LIMITATION

Counting must include tactical details such as promotion square control and whether the new queen gives check.

DECISION HIERARCHY

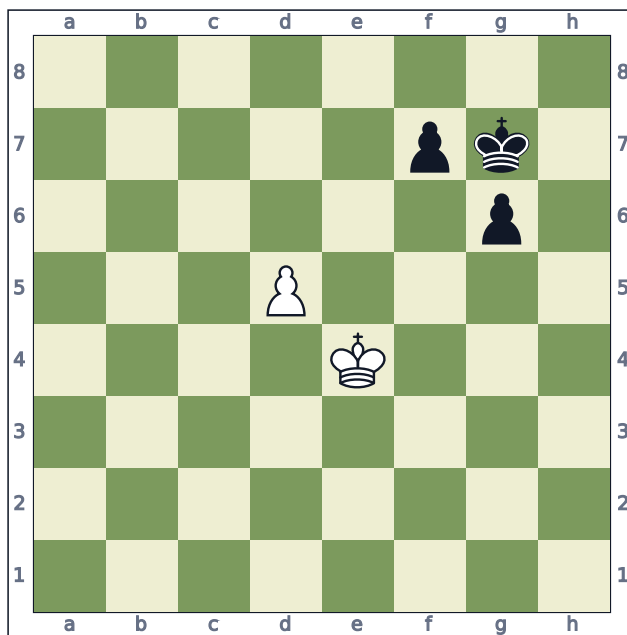
- 1. Legal and tactical survival: checks, forced mate, and immediate material loss.
- 2. King safety and forcing initiative: whose threats arrive first?
- 3. Material and conversion: what durable gain remains when activity fades?
- 4. Piece activity, structure, space, and long-term plans.
- 5. The general principle as a tiebreaker between otherwise sound choices.

RED-FLAG TEST

Before following the rule, look for one changed defender, an exposed king, an intermediate move, a pawn break, or a tactical capture. If any exists, calculate it first. A principle explains typical positions; the board decides this position.

WORKED THINKING MODEL

How a strong player uses it



Illustrative endgame position - analyze the idea, not a memorized move

1. Read the position

Start with conversion and tempo: identify the feature that will remain important after one move.

2. Find the trigger

Look for these signs: Passed pawns on both wings; kings cannot stop everything; exchanges lead to a race.

3. Produce candidates

Include one forcing move, one move guided by the principle, and one move that prevents the opponent's best idea.

4. Verify

Calculate checks, captures, and threats for the opponent. Try to refute your preferred candidate.

5. Evaluate

Use the mechanism as the explanation, not the proof: Pawn races are concrete. Promotion with check, a diagonal king route, or one reserve tempo can reverse the result.

Note: the board is an illustration for the thinking process, not a claim that one move is forced. Set it up and compare multiple legal candidates.

TRAINING AND RETENTION

Make it automatic

PRIMARY DRILL

Solve twenty pawn races without moving pieces, then verify on a board and record every counting error.

SELF-TEST

- Explain the principle without using its title.
- Name two position signals that make it relevant.
- Give one reason the principle works.
- State the exception that most often causes misuse.
- Describe the calculation check required before applying it.
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ANSWER RUBRIC

A strong answer connects the rule - Count moves to promotion, checks, captures, and king entry rather than relying on visual intuition. - to its mechanism, names a trigger from a real position, and explains this limitation: Counting must include tactical details such as promotion square control and whether the new queen gives check.

SPACED PRACTICE LOOP

Review today, then after 3, 7, 14, and 30 days. On each review, retrieve the idea before opening the PDF. Retire it only when you can recognize the trigger in a fresh position and explain an exception.